

★ WEEK 5 · PART 2 · PRACTICE

The GARCH Model, Worked by Hand

Add a Memory of Variance, Then Forecast the Storm

The GARCH(1,1) recursion, persistence, and the forecast

Part 2's workhorse, with the arithmetic shown. We run the GARCH(1,1) recursion - baseline, reaction, and a memory of yesterday's variance - then read persistence and the volatility half-life, and forecast risk forward as it mean-reverts to its long-run level.

2P

PART 2 · APPLICATIONS

Memory Makes Volatility Persist

Case A: run the GARCH(1,1) recursion, then compute persistence, the long-run variance, and the volatility half-life.
Case B: forecast volatility several steps ahead as it reverts to its long-run level, and meet the leverage and fat-tail extensions. Then a Claude Code lab.

What Part 2 Told Us

Now we apply it - with numbers

1

GARCH adds a memory of variance

To the ARCH reaction it adds yesterday's variance itself, so one beta term replaces many ARCH lags - far more parsimonious.

2

Alpha plus beta is persistence

Their sum tells you how long volatility shocks last. Near one means highly persistent, with a slow-decaying half-life.

3

Volatility mean-reverts, so it forecasts

The forecast pulls back to the long-run variance at the persistence rate - tomorrow's risk, and the whole horizon beyond.

Everything here is a calculator exercise: one recursion, one half-life, one forecast path.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the three GARCH parameters.
- Run the recursion, or forecast forward.
- Then read persistence and risk.

What you should watch

- How the memory term makes volatility persist.
- How persistence sets the half-life.
- How the forecast reverts to the long run.

The GARCH(1,1) Recursion

Baseline, reaction, and memory

GARCH(1,1) runs the world's risk desks with just three numbers.

Today's variance is a baseline, plus a reaction to yesterday's squared shock, plus a memory of yesterday's variance. We run the recursion once, then read the persistence, the long-run variance, and the volatility half-life it implies

The model

- Baseline omega 0.02.
- Reaction alpha 0.09, memory beta 0.90.
- Yesterday: shock -2.0, variance 1.0.

What we'll do by hand

- Run the recursion for today's variance.
- Add up persistence.
- Compute the long-run level and half-life.

The GARCH(1,1) Recipe

Three terms, one variance

$$\sigma_t^2 = \underbrace{\omega}_{\text{baseline}} + \underbrace{\alpha \varepsilon_{t-1}^2}_{\text{reaction}} + \underbrace{\beta \sigma_{t-1}^2}_{\text{memory of variance}} \quad (\text{one } \beta \text{ replaces many ARCH lags})$$

The three terms

- A constant baseline the variance drifts around.
- A reaction to the latest squared shock.
- A memory of yesterday's variance.

Why the memory helps

- One beta term captures a long shock history.
- It replaces many ARCH lags.
- Three parameters do the work of dozens.

Run the Recursion

Baseline, reaction, memory

$$\sigma_t^2 = 0.02 + 0.09(-2.0)^2 + 0.90(1.0) = \underbrace{0.02}_{\text{base}} + \underbrace{0.36}_{\text{react}} + \underbrace{0.90}_{\text{memory}} = 1.28$$

The three pieces

- Baseline contributes 0.02.
- The reaction to the -2.0 shock adds 0.36.
- The memory carries 0.90 forward.

The total

- They sum to a variance of 1.28.
- The memory term dominates.
- That's what makes volatility persist.

Persistence and Half-Life

How long storms last

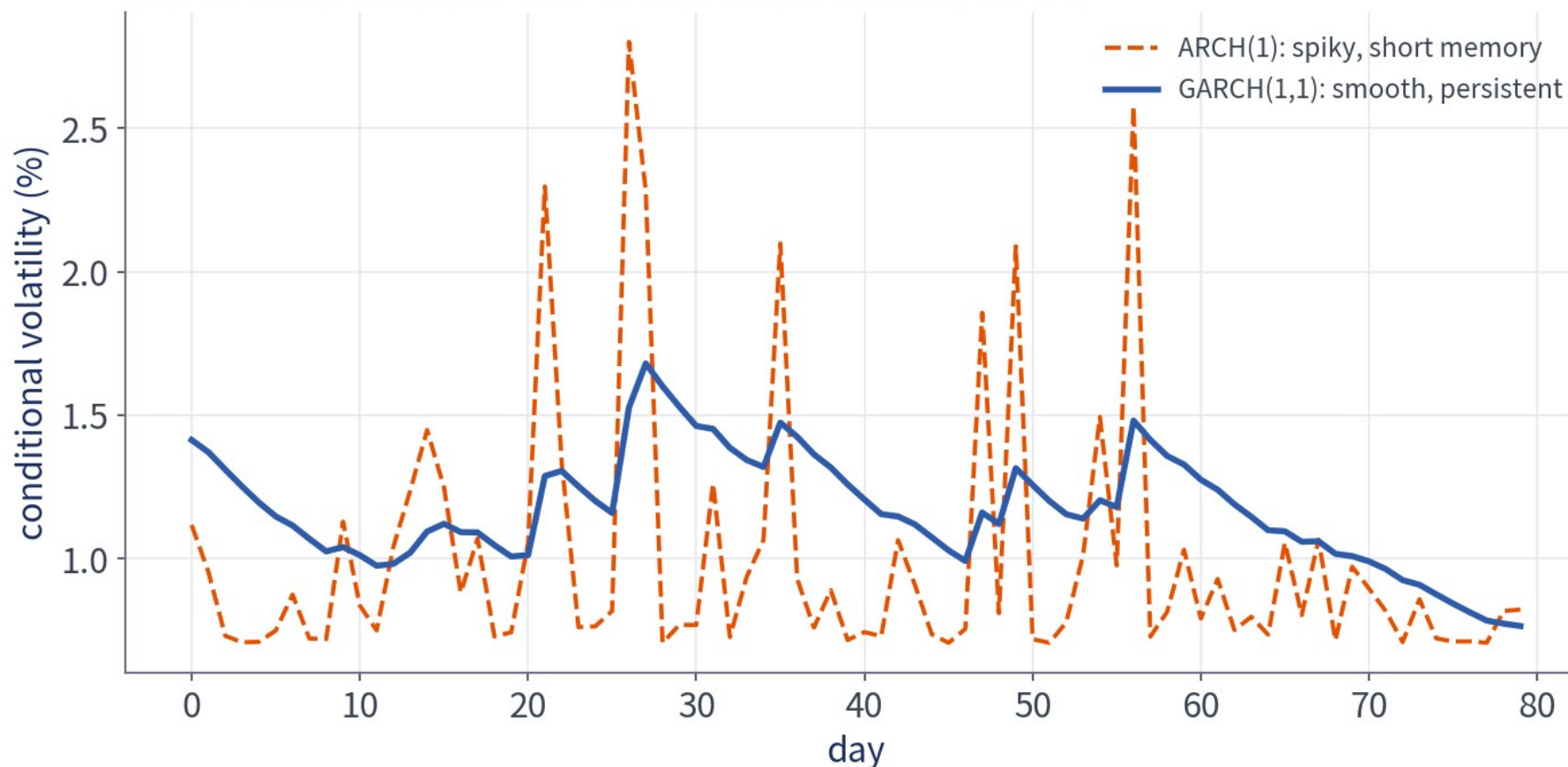
$$\alpha + \beta = 0.99 \text{ (persistence),} \quad \bar{\sigma}^2 = \frac{\omega}{1 - 0.99} = \mathbf{2.0}$$
$$\text{half-life} = \frac{\ln 0.5}{\ln 0.99} = \mathbf{69} \text{ days}$$

Alpha plus beta is 0.99 - very persistent. The long-run variance is 2.0, and the volatility half-life is 69 days: a shock takes about ten weeks to fade halfway. Typical of equity markets.

Memory Smooths the Storm

On the same shocks, ARCH (dashed) spikes and drops sharply - it has no memory. GARCH (solid) rises and then decays gradually, because yesterday's variance carries into today. That persistence is why GARCH matches the slow-fading volatility of real markets with only three parameters

Case A: GARCH's memory term smooths and prolongs the volatility response



Why GARCH Won

Parsimony and persistence

What the memory buys

- A long volatility history in one beta.
- Persistence markets actually display.
- Three parameters instead of many lags.

Why it's the workhorse

- Simple enough to estimate reliably.
- Rich enough to fit equity volatility.
- The default on every risk desk.

Forecasting the Storm

Risk over the whole horizon

Because volatility mean-reverts, GARCH forecasts it. Today's variance sits below its long-run level, so the forecast climbs back toward it at the persistence rate - fast if persistence is low, slow if it's near one. We compute the forecast at several horizons, then meet the leverage and fat-tail extensions that make it realistic

The starting point

- Today's variance is 1.28.
- The long-run level is 2.0.
- Persistence is 0.99 - slow reversion.

What we'll do by hand

- Forecast at one, five, and twenty days.
- Watch it climb toward the long run.
- Then weigh the extensions.

The Multi-Step Forecast

Mean reversion to the long run

$$\mathbb{E}[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^h(\sigma_t^2 - \bar{\sigma}^2) \quad (\text{reverts to the long-run level})$$

How it works

- Start from today's variance.
- Decay the gap to the long run.
- At the persistence rate each step.

What it implies

- Low persistence: fast reversion.
- High persistence: a slow, gentle climb.
- The long run is the anchor.

The Forecast Path

Climbing back to the long run

horizon	variance forecast	volatility
$h = 1$	1.29	1.14%
$h = 5$	1.32	1.15%
$h = 20$	1.41	1.19%
$h = 100$	1.74	1.32%

From today's 1.28, the forecast inches up - 1.29, 1.32, 1.41, 1.74 - toward the long-run 2.0. The climb is slow because persistence is 0.99; even at a hundred days the forecast hasn't fully arrived.

Two Realistic Extensions

Leverage and fat tails

leverage: bad news raises vol more (GJR); fat tails: Student- t innovations, not normal

The leverage effect

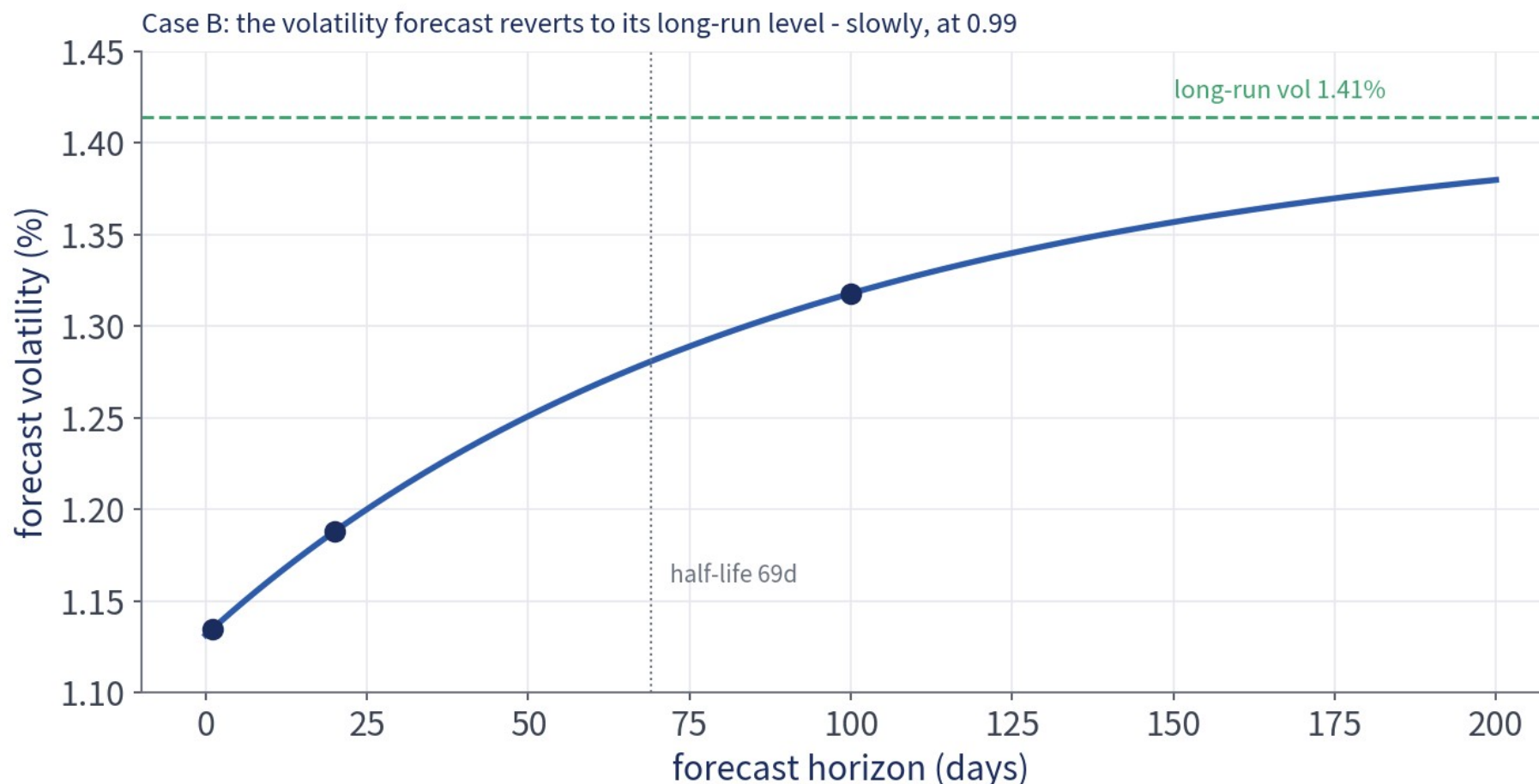
- Bad news raises volatility more than good.
- Plain GARCH treats them the same.
- GJR-GARCH adds an asymmetry term.

Fat tails

- Normal shocks are too thin-tailed for returns.
- Real crashes happen more often than normal says.
- Student- t innovations fix the kurtosis.

The Volatility Term Structure

Starting below its long-run level, the forecast volatility climbs slowly toward the dashed 1.41% line. The half-life of 69 days marks where it has closed half the gap. High persistence makes this a long, gentle curve - today's calm lingers for weeks before fully normalizing



Risk Has a Horizon

And a realistic shape

Why the forecast matters

- Risk over one day differs from over a month.
- The term structure feeds directly into VaR.
- Persistence sets how long calm or storm lasts.

Why the extensions matter

- Leverage captures the crash asymmetry.
- Fat tails price the true chance of extremes.
- Together they make the risk numbers honest.

The GARCH Workhorse

Why it works the risk best

GARCH(1,1) earns its place with three numbers. The baseline sets the long-run level, the reaction responds to the latest shock, and the memory makes volatility persist. Persistence gives the half-life; mean reversion gives the forecast. Add leverage and fat tails, and you have a model that captures how real market risk clusters,

The recursion (Case A)

- Variance 1.28: base + react + memory.
- Persistence 0.99, half-life 69 days.
- Long-run variance 2.0.

isi

The forecast (Case B)

- Climbs 1.29 to 1.74 over 100 days.
- Reverts to the long-run 2.0.
- Leverage and fat tails refine it.

Reproduce the Recursion and the Forecast

Direct Claude Code to check your arithmetic

GOAL Run GARCH(1,1), read persistence and half-life, then forecast the variance.

[INPUT] Prompt · Claude Code

```
Write and run part2_practice.py that, with
omega=0.02, alpha=0.09, beta=0.90:
CASE A - recursion:
    * with shock -2.0 and prev variance 1.0, print
      today's variance broken into its three terms;
    * print persistence, unconditional variance,
      and the volatility half-life.
CASE B - forecast:
    * from variance 1.28, print the h-step forecast
      for h = 1, 5, 20, 100.
Then plot ARCH vs GARCH, and the forecast path.
```

Check against the slides

- variance 1.28
($0.02 + 0.36 + 0.90$).
- persistence 0.99, half-life
69d.
- unconditional variance 2.0.
- forecast 1.29 -> 1.74.

Deliverable: part2_practice.py whose printed numbers match the recursion and the forecast path.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does the beta term add over ARCH?

A memory of past variance - one term for many lags

Q2 What does alpha plus beta measure?

Persistence - how long volatility shocks last

Q3 How does GARCH forecast volatility?

It reverts to the long-run level at the persistence rate

All three come from the recursion, the persistence, and the multi-step forecast.

Three Pitfalls to Avoid

Where GARCH trips people

Common mistakes

- "Persistence near one is fine to ignore."
- "Normal shocks capture crash risk."
- "Good and bad news move volatility equally."

The reality

- Near one means very slow reversion - long memory.
- Use Student-t: real tails are fatter than normal.
- Bad news hits harder - use a leverage term.

Two Calculations to Remember

Part 2 applications in one screen

1

Three terms build the variance

The GARCH(1,1) recursion took a baseline of 0.02, a reaction of 0.36 to the -2.0 shock, and a memory of 0.90 to a variance of 1.28. Persistence was 0.99, the long-run variance 2.0, and the volatility half-life 69 days - a slow-fading storm.

2

Volatility mean-reverts, so it forecasts

From today's 1.28, the forecast climbed to 1.29, 1.32, 1.41, and 1.74 over one to a hundred days, reverting toward the long-run 2.0. Leverage and fat-tail extensions then make the shape and the extremes realistic.

3

Three numbers run the risk desks

Baseline, reaction, and memory - plus persistence for the half-life and mean reversion for the forecast. That's why GARCH(1,1) is the volatility workhorse of modern finance.

Next — Part 3 practice: turning a volatility forecast into Value-at-Risk and a volatility-timed pairs trade.